B.Voc.(Food Processing and Preservation)

Session 2016-17

Semester I

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-101	Communication Skills	Generic	Theory	20	80	-	**-**	**6**
*GEN-102	Fundamentals of Information Technology	Generic	Theory	20	80	**-**	**-**	**6**
FPP103	Bakery and Confectionary	Skill	Theory& Practical	10	40	10	40	6
FPP 104	Dairy Science & Technology	Skill	Theory& Practical	10	40	10	40	6
FPP105	Food Quality Assurance	Skill	Theory & Practical	10	40	10	40	6

Semester II

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-201	Soft Skills and Personality Development	Generic	Theory	20	80	-	**-**	**6**
GC 202	Business Ethics	Generic	Theory	20	80	**-**	**-**	**6**
FPP 203	Food Packaging Technology	Skill	Theory & Practical	10	40	10	40	6
FPP 204	Industrial Safety, Hazards & Prevention	Skill	Theory & Practical	10	40	10	40	6
FPP 205	Food Plant Layout	Skill	Theory & Practical	10	40	10	40	6

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce These credits will be evaluated in Semester VI

Semester III

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-301	Value Education And Human Rights	Generic	Theory	20	80	-	**-**	**6**
GC 302	E-Commerce	Generic	Theory	20	80	**-**	**-**	**6**
FPP - 303	Food Fermentation & Beverage Technology	Skill	Theory& Practical	10	40	10	40	6
FPP - 304	Milk &Dairy Food Product Development	Skill	Theory& Practical	10	40	10	40	6
FPP - 305	Nutraceuticals & Probiotics	Skill	Theory & Practical	10	40	10	40	6

Semester IV

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-401	Environmental Studies	Generic	Theory	20	80	-	**-**	**6**
GC 402	Project Management	Generic	Theory	20	80	**-**	**-**	**6**
FPP - 403	Meat, Poultry & Fish Processing	Skill	Theory & Practical	10	40	10	40	6
FPP - 404	Food Engineering	Skill	Practical	10	40	10	40	6
FPP - 405	Seminars	Skill	Practical	-	-	100	-	6
***Refers to all Generic Components common to all B.Voc. Courses Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce These credits will be evaluated in Semester VI**								

Semester V

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-501	Critical Thinking And Elementary Statistics	Generic	Theory	20	80	-	-	**6**
GC 502	Introduction To Research Methodology AndReport Writing	Generic	Theory	20	80	-	-	**6**
FPP - 503	Food Biotechnology	Skill	Theory& Practical	10	40	10	40	6
FPP - 504	Cereal and Pulse Technology	Skill	Theory& Practical	10	40	10	40	6
FPP - 505	Bakery & Confectionary Technology	Skill	Theory & Practical	10	40	10	40	6

Semester VI

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-601	Entrepreneurship Development Programme	Generic	Theory	20	80	-	-	**6**
GC 602	Total Quality Management	Generic	Theory	20	80	-	-	**6**
FPP - 603	Industrial Training	Skill	-	-	-	-	-	Non CRedit

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce These credits will be evaluated in Semester VI

Semester I

Code	Skill Component	Job Description
FPP103	Bakery and Confectionary	**Baking Technician:** Responsible for baking of products and maintaining their quality
FPP 104	Dairy Science & Technology	**Dairy Technologist:** To work with dairy products like milk, butter, yogurt, cheese. To prepare and preserve high quality dairy products
FPP105	Food Quality Assurance	**Food quality assurance manager**: To determine and establish procedures and standards for food production, preservation and quality management. Is responsible for implementing and ensuring that food products produced meet standards set by both the organisation and regulatory authorities.

Semester II

Code	Skill Component	Job Role
FPP203	Food Packaging Technology	**Food Packaging technician:**performs various packaging functions and handles all categories of packaging such as primary, secondary and tertiary packaging for food products.
FPP 204	Industrial Safety, Hazards & Prevention	To evaluate values of industrial safety and hygiene. Is responsible for monitoring and assessing hazardous hazardous and unsafe situations and developing measures to assure workers safety
FPP205	Food Plant Layout	**Food Plant layout officer:** Managing and updating food plant layout time to time as per business policies and safety standards.

Semester III

Code	Skill Component	Job role
FPP 303	Food Fermentation & Beverage Technology	**Fermentation microbiologists** work in academic, basic and applied research laboratories, where they study the fermentation processes of microorganisms like bacteria, yeasts, fungi and mold. Develop and improve processes for making beer, wine, cheese and other products
FPP 304	Milk &Dairy Food Product Development	**Dairy scientist**: QA Management, *milk quality* testing, *dairy products* testing Marketing Executive in dairy industry.
FPP 305	Nutraceuticals and probiotics	Working with specific strains of microorganisms and developing new

		supplements and drinks to improve the overall health of human beings especially disgestive system

Semester IV

Code	Skill Component	Job role
FPP 303	Meat, Poultry & Fish Processing	To work in an abattoir, wholesale meat factories and meat processing plants. A meat processor is involved with the production of meat and poultry products.
FPP 304	Food Engineering	***Food engineers*** utilize the latest innovative technology to process, package, preserve and improve food products. This type of engineer works in combination with the agricultural and food processing industry to ensure food safety, supply, nutrition and value. To develop and design new food equipments.
FPP 305	Seminars	To improve the communicating and presentation skills of the individual.

Semester V

Code	Skill Component	Job role
FPP 503	Food Biotechnology	**Food** biotechnologists **work** in the **food** and drink industry developing a wide range of products and making sure they are safe for consumers. Developing GM foods by using techniques of biotechnology including genetic engineering etc.
FPP 504	Cereal and Pulse Technology	**A Cereal and Pulse Processing Technician** is responsible for milling various types of cereals and pulses (red gram, black gram, bengal gram, green gram, green peas, etc.) through processes such as cleaning,, destoning, sorting etc.
FPP 505	Bakery & Confectionary Technology	**Baking Technician**: Responsible for baking of products and maintaining their quality. Developing new techniques for production of better and high quality bakery items

Semester VI

Code	Skill Component	Job role
FPP 603	Industrial Training	

Semester I

FPP 103: BAKERY AND CONFECTIONERY

Course Objectives:

• To understand the composition of different ingredients used in Baking Industry.
• To know the methods of processing and preservation of foods.
• To identify the microorganisms of food commodities of plant and animal origin.
• To learn about Food borne diseases and microorganisms.

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks.

Unit I

Introduction, Types and processing mechanism:Flour: Structure, Types, Milling and Storage. **Rice**: Types, forms and processing. **Milk, Butter and Cream**: Types, Processing,Byproducts, Storage. **Cheese**: Classification and uses. **Raising and Thickening Agents**: Classification and types.**Egg:** Structure and nutritional composition

Unit II

Conventional Methods Of Preservation: Thermal mode of preservation – Pasteurization,sterilization and appertization – Heat resistance of microorganisms and their spores –bacteria and their spores –fungi and their spores –yeast and their spores - spoilage of canned foods and types of spoiled cans – aseptic packaging - Low-temperature storage. Non-thermal methods ofpreservation : High pressure processing – Pascalisation.

. Unit III

Introduction to Baking and Confectionary process methods: Methods of Bread Making: Straight dough method, Sponge dough method, Cake making: Sugar batter, Flour batter method, Pastry Making: Principle and types. Biscuit making.

Unit IV

Bread Diseases: Rope and Mould, Factors responsible for it and preventive measures.
Faults and Remedies: Basic reasons and suggested remedies: Bread, Cake and Biscuits.

Text books/ References:

1. Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2ndEdition.
2. Frasier.1987. Food Microbiology.Tata McGraw-Hill Education
3. Bakery-I, Student handbook and practical manual, published by CBSE.
4. Ornamental Confectionary and the Art of Baking in all its Branches by Herman Hueg
5. Bread: A baker's book of techniques and recipes by Jeffrey Hamelman.
6. A professional Text to bakery and confectionary by John Kingslee

Practical based onFPP103

Time: 3 hours

1. Preparation of White Bread.
2. Principle and preparation of Fruit cake.
3. Studying the effect of temperature on process of biscuit making.
4. To make garlic bread and sensory analysis.
5. To do icing on the cake.
6. Isolation of microbes from spoiled bread.
7. To study effect of pasteurization on milk
8. Methylene blue reduction test in milk

FPP 104: DAIRY SCIENCE & TECHNOLOGY

Course Objectives:

• To create interest among students about different aspects of dairy industry.
• To study quality standards and production of various types of milk and milk products.
•To study the role of dairy farming in Indian economy

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks.

Unit I

Milk: Physicochemical properties, quality of milk, grading, dye detection test, platform test, sensory evaluation, milk borne diseases. Spoilage of milk: causes and prevention.

Types of market milk: TM, DTM, SPM, RRM and fortified milk. Effluent treatment of dairy plant disposal.

Unit II

Dairy Animals: Care and management of different dairy animals. New borns, milch, pregnant animals, bull and sick animals. Feed formulation; Nutrient requirement of different dairy animals Common diseases: bacterial, viral, fungal, prevention and their control (Vaccination, deworming). Development of transgenic dairy animals.

Unit III

Composition, standards, manufacturing: Process, equipments and defects during manufacturing and storage of dairy products and by products (Cream, paneer, yogurt, milk powder- skimmed milk and whitener, casein, whey concentrate, lactose, ghee residue).

Unit IV

Dairy development in India: Present dairying- Present status, future prospective and its role in Indian economy. Important government initiative (operation Blood). Role of dairy development organizations (NDRI, IVRI ,Amul) in dairy development

Text books/ References:

1. A textbook of animal husbandry, G.C. Banerjee, IBH Publication
2. Farm animal management and poultry production, Shastri Thomas and Singh, Vikas Publishers
3. Indian Dairy products, K.T. Acharya
4. Fliud milk industry, J.S Handerson, A.V.I Publishing Company, USA1
5. Milk Hygiene in milk production processingand distribution, F.A.O Publication.

Practical based onFPP 104

1. Gerber fat test for milk.
2. Sampling of milk and milk products for microbiological analysis
3. Platform test for milk analysis.
4.Proximate analysis of feed: Dry matter, nitrogen, crude fiber and total ash.
5.Visit and study a nearby milk union/ dairy and prepare a checklist of problems in procurement and milk distribution
6. Listing of quality control agencies at national and international levels.
7. Preparation of flavored milk: TM and DTM

FPP105: FOOD QUALITY ASSURANCE

Course Objectives:

• To understand the different principles of food quality control.
• To assess the food quality assurance of bakery products.

Instructions for paper setters

- The syllabus of this paper has been divided into four units.

- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks.

Unit 1

Objectives, importance and functions of quality control.Principles of food quality control and quality assurance, quality control and assessment in food materials used in baking industry and finished bakery products.

Unit II

Total quality management (TQM) - good manufacturing practices, good hygienic practices, good lab practices, general awareness and role of management practices in quality control.

Microbial quality control: determination of microorganisms in foods by cultural, microscopic, physical, chemical, immunological and bioassay methods

Unit III

Food regulations, grade and standards, concepts of Codex Almentarious, HACCP, USFDA, ISO 9000 series etc. Food laws and standards, Food standards and safety Act: salient provisions and prospects, role of various national and international agencies.

Unit IV

Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituents of bakery products

Sensory quality evaluation: Introduction, methods, panel screening, selection methods, Sensory and instrumental analysis in quality control,

Practical based onFPP 105

1 Techniques of quality assessment of different natural and processed foods.
2 Identification and ranking of food product attributes
3 Identification and ranking of food product attributes.
4. Sensory methods for measuring foodquality assessment of raw materials used and processed bakery products
5. Instrumental methods for measuring foodquality assessment of raw materials used and processed bakery products
6. Study of cleaning and sanitizers used in pre and post-operative processes in bakery industry
7. Documentation of details of baking ingredients, process and finished products used in baking industry.

Text books/ References:

1. InteazAlli (2003). Food Quality Assurance Principles and Practices.CRC Press.

2. J Andres Vasconcellos (2003) Quality Assurance for the Food Industry. A Practical Approach.CRC Press

Semester II

Paper Title: GC 202 - BUSINESS ETHICS

Objective: The objective of this paper is to familiarize the students with the importance of ethics in business and understanding of issues related to corporate social responsibility and corporate governance.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Unit 1

Business Ethics: Meaning and Concept, Principles of Business Ethics, Characteristics of Ethical Organisations, Theories of Business Ethics.

Unit-II

Globalization and Business Ethics, Stakeholder's Protection, Corporate Governance and Business Ethics. Ethical Issues in Indian Business.

Unit - III

Professional Values:Objectives, Morals, Values, Ethics, Integrity, Work ethics, Service learning, Virtues, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Challenges in the work place, Spirituality.

Unit - IV

Corporate Social Responsibility**:** Social Responsibility of business with respect to different stakeholders, Arguments for and against social responsibility of business, Social Audit,Corporate Social Responsibility and Corporate Governance.

Suggested Readings:

1. J.P. Sharma, *Corporate Governance, Business Ethics & CSR,* Ane Books Pvt. Ltd., Nee Delhi.
2. Andrew Crane, Dirk Matten, *Business Ethics,* Oxford University Press, New Delhi.
3. Daniel Albuquerque, *Business Ethics, Principles and Practices (Indian Edition),* Oxford University Press, New Delhi
4. Fr. Floriano C. Roa, *Business Ethics and Social Responsibility*, Rexestore.
5. O. C. Ferrell, John Fraedrich, Linda Ferrell, *Business Ethics: Ethical Decision Making & Cases, Cengagae Learning*
6. Michael Blowfield, Alan Murray, *Corporate Responsibility – A Critical Introduction,* Oxford University Press, New Delhi.

FPP 203: FOOD PACKAGING TECHNOLOGY

Course Objective:

• To enable the students to understand about packaging and packaging materials, interaction of food items with packaging materials

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks.

Unit I

Introduction to Food Packaging: Functions of packaging, Effect of environmental factors on quality of food.

Estimation of shelf life, analysis of storage requirement, accelerated storage studies: Vacuum and Inert Gas Packaging: Tests on packaging materials, Mechanical strength (Tension, notch and tearing strengths), Gas and water vapor transmission rates.

Unit II

Metal Cans as Packaging material: Types of Metallic Cans - Tin cans and Aluminum cans. Specialty of Open top sanitary cans, Lacquers and their use, three piece cans and two piece cans, Aerosol Cans.

Introduction to Canning operations – Can Reformer,Flanger, Seaming, Can closures, Glass jars and Bottles in food packaging, Design features and applications, Sterilization of jars and bottles.

Unit III

Flexible Films Packaging: Formation of Films and pouches, Plastics used and their Specific applications.

Copolymers their applications.Co-extruded films and Laminates.Rigid and Semi rigid plastic packaging methods. Extrusion –Retort pouch packaging. Laminated Paper board Cartons, Fiber Board and Corrugated Card Board packaging and their applications.

Unit IV

Filling And Sealing Operations For Various Types of Packages: Closing and sealing of rigid plastic containers. Filling and sealing of Flexible plastic containers, Seal types, hot wire sealing, hot bar sealing and impulse sealing.

Active packaging, Moisture control, CO_2 and Oxygen scavenging, modified atmosphere packaging –principles, applications. Permeability of gases in packs.

Text Books/ References.

1. Coles, R., Dowell, D.M., Kirwan, J. "Food PackagingTechnology".BlackWellPublishing Ltd.ISBN-084939788X, 2003.
2. NIIR Board: Food Packaging Technology Handbook National Institute of Industrial Research, New Delhi (2004)
3. PirengerO.G.andA.L.Baver: Plastic Packaging Materials for Food Wiley VCH, GmbH, Germany (2000).

Practical based onFPP 203

1. Demonstration of filling process.
2. To study quality and strength of packaging materials.
3. Demonstration. of measurement of cartons' dimensions as per organizational standards.
4. Demonstration of measurement of dimensions of bottle mouth, cans and their caps.
5. Demonstration of sealing processes used in food industry.

FPP204: INDUSTRIAL SAFETY, HAZARDS & PREVENTION

Course Objectives:

• To create awareness about health hazards of industrial substances.
• To evaluate the threshold value of industrial hygiene and safety.

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks.

UNIT-I

Origin of process hazards, Law Codes, Standards, Properties of Chemicals, and Health hazards of industrial substances.

UNIT-II

Microbial Contaminants associated with food–Bacteria, viruses, fungus, molds and yeast.
Factors affecting the growth of microbes in food. General Microbiological Methods of enumeration and isolation of food related microbes.
Methods of Preservation: Pasteurization,sterilization and appertization

UNIT-III

Toxicology :Toxic materials and their properties, effect of dose and exposure time, relationship and predictive models for response, Threshold value and its definitions, material safety data sheets, industrial hygiene evaluation.

UNIT-IV

Propagation of fire and effect of environmental factors, ventilation, dispersion, purifying and sprinkling, safety and relief valves.

Text books/ References:

1. Crawl D.A. And Louver J.A., 2001 "Chemical Process Safety Fundamentals With Applications" Prentice Hall
2. Wentz, C.A, 2001. "Safety Health And Environmental Protection," McGraw Hill
3. Smith, B.D, 2001 "Design Of Equilibrium State Process, "McGraw Hill.

Practical based onFPP 204

Time: 3 hours

1. Study of morphology of bacteria, yeast and fungi
2. Methods of sterilization and preparation of media
3. Gram staining, negative and lactophenol staining

4. Methods of pure culture techniques for bacteria
5. Enumeration and isolation of bacteria and fungi from water/milk and contaminated food
6. Demonstration of different safety aspects and maintenance of material safety data sheets followed in food industry.

FPP205: FOOD PLANT LAYOUT

Course objective:

- Exposure of the students to the basic setup of a Food industry
- To make them conversant with the machinery and equipments in a food industry

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks.

Unit I

Basic concepts of industrial plant layout and design with special reference to food packaging industries. Application of HACCP concept, ISO, FPO & MPO requirements in food plant layout and design.

Unit II

Design consideration for location of food plants. Basic understanding of equipment layout and ventilation in food processingplants.Preparation of flow sheets for material movement and utility consumption in food plants.

Unit III

Plant layout and design of bakery and biscuit industries.Plant layout and design of fruits and vegetables processing industries including beverages.

Unit IV

Plant layout and design of milk and milk products. Miscellaneous aspects of plant layout and design like provision for waste disposal, safety arrangements etc.

Text Books/ References.

1. Manufacturing Facilities Design and Material Handling by Fred E. Meyers, and Matthew P. Stephens, 3rd Edition, Pearson Prentice Hall, 2000.
2. James M Moore, "Plant Layout and Design", McMillan & Co., (1959)
3. J M Apple, "Plant layout and Material handling", John Willey & Sons, (1977)

Practical based onFPP 205 **Time: 3 hours**

1. Industrial visit and report making.

Semester III

GC 302 –E-COMMERCE

Objective : The objective of this paper is to provide fundamental knowledge to the students about E-Commerce so that they can better perform in any area of operation and can excel in the field of commerce with IT specialization.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Electronic Commerce Framework, Basics and Tools of E-Commerce, Comparison of Web-based with Traditional Business; Growth of E-Commerce – Present and potential. E-Business: Meaning, Importance, Models Based on the Relationships of Transacting Parties (B2B, B2C, C2C and C2B), Present Status of E-Commerce in India.

UNIT II

Changing Structure of Organization – The Impact of E-Commerce on supply chain management and customer relation, Socio-Economic Impacts of E-Commerce. Electronic Payment System: Types of Payment System — E-Cash and Currency Servers, E-Cheques, Credit Cards, Smart Cards, Electronic Purses and Debit Cards , Risk and Electronic payment Systems.

UNIT-III

Electronic Data Interchange (EDI): EDI and its applications in business; Legal, Security and Privacy issues in EDI, EDI standardization; Internet based EDI, Internal Information Systems, Work-flow Automation and Coordination, Types of Digital Documents, Digital Signatures.

UNIT-IV

Enterprise Resource Planning: The Emergence of ERP Systems, Business benefits of ERP, ERP Modules and design alternatives, challenges of ERP implementation, Business Process Re-engineering, ERP system development process. Regulatory aspects of E-commerce.

Books Recommended:

- 'Electronic Commerce, A Manager's Guide to E-Business' by Prag Diwan and Sushil Sharma, Vanity Books International, Delhi.
- 'Frontiers of Eletronic Commerce' by Ravi Kalakota, and Andrew B.Shinston, Addision Wesley.
- 'Web Commerce Technology Handbook' by Minoli and Minoli, Tata McGraw Hill, New Delhi.
- 'Electronic Commerce' by Gray P.Schneider, Course Technology, Delhi.
- 'E-Commerce-The Cutting Edge of Business' by Kamlesh Bajaj & Debjani Nag, Tata Mc Graw Hill 2000.

'Understanding E-Commerce' by David Kosiur, Microsoft Press. 1997

FPP -303: FOOD FERMENTATION & BEVERAGE TECHNOLOGY

.**Course Objectives**:

• To understand the different types of fermentation technologies used in making fermented.

Instructions for paper setters:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks

Unit I

Introduction to fermentation: Rate of microbial growth and death.

Fermentation kinetics, Fermenter design, operation measurement and control and types of fermenters, Batch/continuous fermentation, scale up in fermentation, Principle and use of biosensors.

Food fermentations; Lactic acid fermentation of milk, vegetable, cereals and meat: Alcoholic fermentation of fruit juices, sugar and starch substrates; Leavening and baking process.

Unit II

Mixed fermentation in cereals, legumes and milk; Role of fermentation in processing of tea, coffee and cocoa;

Microbial polysaccharides production in foods; Indigenous fermented foods like Dahi, Rabri, Idli, Dosa, Jalebi, etc.

Unit III

Production of vitamins, amino acids, organic acids, enzymes and antibiotics, alcohols.

Brewing technology of alcoholic beverages- wine, cider, brandy, mead, perry, toddy, sake.

Unit IV

Beverages : Definition, types, importance of beverages in our diets, soft drinks and hard drinks, treatment of water for food industry.

Technology of soft drinks:Carbonated and non-carbonated beverages.

Whey beverages and utilization of whey in development of fortified drinks, use of low calorie sweeteners in beverages.

Text books/ References:

1. Robert W. Hutkins. 2006. Microbiology and Technology of Fermented Foods. Wiley-Blackwell

2.Y H Hui and E Ozgul. Evranuz. 2012. Handbook of Plant Based Fermented Food and Beverage Technology. CRC Press.

Practicals based on FPP -303

1 Lactic acid fermentation of milk and vegetables
2 Alcoholic fermentation of fruit juices
3 Acetic acid and alkaline fermentation; Baker's yeast activity
4 Barley steeping, germination, malting, mashing and brewing of malt
5 Preparation of whey beverages,
6 Chemical and Microbiological analysis of water for mineral content, turbidity, reaction, hardness, plate count and coliform counts
7 Millipore filtration; Treatment of water for food industry.

FPP-304: MILK AND DAIRY FOOD PRODUCTS DEVELOPMENT

Course Objectives

To enable thestudents to understand the various dairy products and their preparation.

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks

Unit 1

Dairy industry in India & its future prospects.
Composition and properties of milk, liquid milk handling operations: Collection and Testing; Filtration and Clarification, Skimming,
Standardization; Homogenization, Pasteurization (LTHT, HTST, UHT)
Sterilization, Packaging, Storage and distribution of fluid milk;

Unit II

New Concept in milk processing - UHT, Membrane processing, Microwave and irradiation treatments; aseptic packaging.

Microbiology of milk, sources of milk contamination.

Unit III

Technology of indigenous dairy products: Dahi, butter, ghee, channa, paneer etc.
Technology of cream, butter, margarine and ghee manufacture.
Technology of condensed and evaporated milk: Composition, nutritive value, process of manufacture, defects (their causes and prevention).
Technology of yoghurt, Acidophilus milk, Bufgaricus milk, Kumiss, Kefir; Manufacturing of Cheddar, Mozzarella, cottage and processed cheese.
Technology of milk powders (WMP, SMP): Composition, nutritive value, process of manufacture, defects (their causes and prevention), instantization of milk powder.

Unit IV

Milk and milk product standards and legislations in India: Grading of milk and criterion of grading.

Milk adulteration and quality control in dairy industry.By products of dairy industry and their utilization.

New concepts in technology of dairy products, cream powder, sterilized cream, butter spread, butter powder, cheese spread; packaging and distribution of various milk products; Defects in milk products and their remedies; Milk based infant foods; Manufacturing of casein, Caseinate,

National and International Organizations in dairy Industry.

Text Books/References:

1. Sukumar.1983. Outlines of Dairy Technology.De,Oxford University Press.
2. James N.Warner.1989. Principles of Dairy Processing by, Wiley Eastern Ltd.
3. Eckles,Combs; and Macy. 1982. Milk and Milk Products by,Tata McGraw Hill.
4. Anejaet al.2002.Technology of Indian Milk Products. A Dairy India Publication.

Practicals based on FPP-304

1 Platform tests,
2 Determination of SNF, TS Protein,
3 Lactose and ash contents of milk.
4 Layout plan for setting up of milk plant,
5 Preparation andEvaluation of different types of milk and milk products .
6 Preparation and evaluation of butter, ice cream, cheese, yoghurt.

FPP-305: NUTRACEUTICALS & PROBIOTICS

Course Objectives

To enable the students to understand the concepts of nutraceuticals and probiotics.
To enable the students to understand the importance of food for good health.

Instructions to paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks

Unit 1

Concept on nutraceutical:nutraceutical and functional foods, nutraceutical as new dietary ingredients, biological significance of Nutraceuticals, Nutraceuticals and dietary supplement, world market for Nutraceuticals.

Unit II

Nutraceuticals in food and beverage industry: non caretonidterpeniods, flavonoids and other polyphenolic compound, natural pigments (chlorophyll, chlorophyllin, xanthophylls, betacaretonoides).

Unit III

Antioxidants, organic acids terpenoids, alkaloids amino acids. Current technology for production of Nutraceuticals : anti-inflammatory, antivirals and antibacterial, antioxidants, and natural pigments.

Unit IV

Concept and opportunities on probiotics, prebiotic and synbiotic, biological significance and use of probiotics for human, for cattle, poultry and for aquaculture. World market for probiotics current technologies.,

Commercial production, current technology in probiotics production.

Probiotic foods, resistant starch, fructo-oligosaccharides as Probiotic food components, Efficacy of probiotic foods, standards and regulation for probiotic foods.

Practicals based on FPP- 305:

1. Quantitation in Nutraceuticals in fruits and vegetables.
2. Isolation of flavonoids from apple pomace,
3 .Calorimetric quantitation of Nutraceuticals,
4. Separation of pigments by TLC.
5. Separation and quantifications of - pigments by column chromatography.

Text Books/References:

1.SemihOtles. 2013. Probiotics and Prebiotics in Food, Nutrition and Health. CRC Press

2. Leah Coles. 2013. Functional Foods: The Connection between Nutrition, Health, and Food Science. CRC Press
3. Lynnette R. Ferguson. 2013. Nutrigenomics and Nutrigenetics in Functional Foods and Personalized Nutrition. CRC Press

Semester IV

Paper Title: GC 402 - PROJECT MANAGEMENT

Objectives: The objective of this paper is to provide knowledge to students about the essentials of undertaking projects in an organizational environment.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Concept of Project: Meaning, Characteristics, Classification of Projects, Project Life Cycle and its Phases. Project Manager: Roles and Responsibilities, Project Management as a Profession.

UNIT-II

Generating and Screening Ideas – Steps, Monitoring the Environment, Scouting for Project Ideas, Preliminarily Screening, Project Rating Index. Feasibility Studies – Technical, Financial, Economic, Social, Legal and Managerial.

UNIT-III

Project Appraisal Techniques: Objectives, Types and Methods. Project Risks: Meaning, Types, Measurement of Risk,. Decision Tree Analysis (Basic Concepts only).
Project Evaluation – Meaning, Evaluation v/s Appraisal, Objectives, Types of Evaluation, Techniques.

UNIT-IV

Project Organization and Control – Project Network Analysis (Basic concepts of PERT, CPM, Cost and Time Over Run). Project Reporting: Meaning, Purpose, Process, Requirements of a Good Report, Methods, Principles of Good Reporting System.

Books Recommended:

- Project Management - Choudhary – TataMcGraw Hill Pub.
- Project Management: The Managerial Process (Special Indian Edit.) -Clifford F Gray, Oregon State University.
- Projects : Planning, Analysis, Selection, Financing, implementation and Review - Chandra, Prasanna.

FPP-403: MEAT, POULTRY & FISH PROCESSING

Course Objectives:

• To create awareness about the processing of meat, fish and poultry.
• To study the storage and handling techniques of fish and poultry.
• To gain knowledge on processing of meat.
• To study about the byproducts obtained during processing along with their uses.

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks

Unit I

Status and scope of meat industry in India.

Structure and physico-chemical properties of muscle.

Source of meat animals and fishes, importance in national economy;

Meat : Composition and nutritive and value, Slaughtering and dressing of meat animal; Quality evaluation of eviscerated carcasses, environmental and animal production factors that affect meat quality, post mortem changes in meat, rigor mortis.

Unit II

Processing and preservation methods of meat, Aging of meat, meat tenderization-natural and artificial methods, Various color changes in muscle and meat, color stabilization by curing,

Storage and preservation of meat: Chilling, Freezing, Curing, Smoking, Dehydration, Freeze-drying, Irradiation, Canning and Glazing of fish.

Unit III

Cooking, palatability and eating quality of meat, microbial spoilage of meat, Restructured meat products, meat analogs, Meat industry of Fish : Factors affecting quality of fresh fish, fish dressing, chilling, freezing, salting and canning of fish, Manufacturing of fish paste, fish sauces, fish oil, fish protein concentrate, fish meal.

Unit IV

Egg: Structure, composition, nutritive and functional properties.
Quality of egg: Internal quality evaluation, egg canning, egg grading, microbial spoilage of eggs, preservation and storage methods for eggs.
Egg products: Egg powder, boiled egg, scrambled eggs. Packaging and transportation of eggs.
Poultry products: Types, chemical and nutritive value of poultry meat,; grading and packaging of poultry meat.

Text Books/References:

1 R.A. Lawrie, 1988 Meat Science, Pergamon Press.

2. G.J. Mountney.1995. Poultry Products Technology by Taylor & Francis,

3. Parkhurst& Mountney.2012. Poultry Meat and Egg Production.Springer London, Limited, 05-Jul-2012

FPP-404: FOOD ENGINEERING

Course Objectives:

- To create awareness about the food processing.
- To study the various methods of preservation of food.

Instructions to paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks

Unit 1

Extrusion technology: General principles, extrusion process, advantages of extrusion, extrusion equipment, single screw extruders and twin screw extruders, effect of extrusion on food properties, extrusion of starch based foods.
Hydrostatic Pressure Technology principles and application of hydrostatic pressure technology in food industry.

Unit II

Hurdle Technology : Principles, types and application of hurdle technology in dairy products, intermediate moisture foods, fermented products, heated foods and chilled foods.
Membrane Technology : Introduction of membrane, their classification and function, principles of reverse osmosis and ultrafiltration, nanofilteration and microfiltration, applications of membranes in food processing industry, modules for using membrane filters.

Unit III

High intensity pulse electric field (HIPEF): Principles, generation of electric field pulses, application in foods.
Food Irradiation Technology: General aspects of irradiation, ionizing radiation, irradiation process, units, mechanism, advantages and disadvantages of irradiation process,

Unit IV

Ultrasound in food processing and preservation : Introduction, ultrasound instrumentation, ultrasound processing for enhancement of mass transfer, heat transfer and homogenization and emulsification.
Microwave Processing: Microwave, properties, heating mechanism, applications of microwave in food processing, effect of microwave on food nutrient, recent studies.
Immobilized enzyme technology: concept, methods of immobilization and use of immobilized enzymes in food processing industry.

Text Books/References

1 Grahame W. Gould. 1995.New Methods of Food Preservation. Springer.

2 Norman N. Potter. 1986 .Food Irradiation and Microwave Heating. Springer

3 M ShafiurRahman. 2007. Handbook of Food Preservation.

FPP--405: Seminars

Objective: -To make the students conversant with latest happening in the field of Food Processing and Preservation and to improve their communicational skills.

Semester V

Paper Title : GC 502 - INTRODUCTION TO RESEARCH METHODOLOGY AND REPORT WRITING

Objective: The objective of the module is to familiarize the students with basics of research methodology like types of research, data collection, sampling techniques and report writing.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Research - Meaning, Characteristics, Types, Process and Utility,. Research Design - Meaning, Types and Features of good research design. Qualitative and Quantitative research. Importance of review of literature. The research proposal.

UNIT II

Primary and Secondary data. Methods of data collection. Measurement and scaling, Designing of Questionnaire and schedule.

UNIT III

Sampling Design and Sampling procedure. Sampling and non sampling errors. Data analysis and interpretation. Role of computers in research.

UNIT IV

Business Reports and Proposals: Introduction, Meaning a Report? Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals; Case Studies.

Books Recommended :

- 'Research Methods and Techniques' by C.R. Kothari, Eastern Wiley Publications.
- 'Business Research Methods' by A. Bryman, and E. Bell, Oxford University Press, New York.
- 'Research Methodology A Step by Step Guide for Beginners' by R. Kumar, Pearson Education.

FPP-503: FOOD BIOTECHNOLOGY

Course Objectives:

• To create awareness about the various GM foods.
• To study the various prospects of biotechnology in the development of food products.

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks

Unit 1

Introduction to Food biotechnology, basic principles of genetic engineering, Application of genetic to food production improvement of the processing of various crops by genetic engineering, food safety and biotechnology.

Transgenic plants and their contribution to food production enhancement.

Unit II

Methods of molecular cloning, immobilization of microbial cultured plant cells. Bacterial starter culture,

Methods of inoculum andmediumpreparation, slurry processing and product isolation.

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Unit III

Natural antimicrobials for food preservation: Phytoalexins, essential oils and their components, bacteriocins, nisin, pediocinsetc, applications of bacetriocins in food systems. Aflatoxins-production, control and reduction using molecular strategy.

Protein engineering in Food technology - methods, applications of protein engineering (e.g. glucose isomerase, Lactobacillus Beta-galactosidase and peptide antibiotic nisin) single cell protein

Unit IV

Fermented food: origin, scope and development, saukraut, younghurt, cheese, misotempeh, Regulatory and social aspects of biotechnology of foods, production of food flavor and colors.

Bio management of food industry wastes. Organic foods: Concept, scope, certification and regulation.

Nanotechnology in food processing .Golden rice : Concept, characteristics and importance of golden rice.

Text Books/ References.

1.Bains W. 1993, Biotechnology from A to Z, Oxford Univ. Press, Oxford.
2.Crueger, W. and Crueger A. 1984. Biotechnology: A Textbook of Industrial Microbiology. Science Tech. Madison, USA.
3. Joshi, V.K. and Pandey, A. Ed. 1999. Biotechnology.Food Fermentation, (2 Vol. set). Education Publ. New Delhi.
4. Knorr, D. 1982. Food Biotechnology. Marcel Dekker, New York.

Practical based on FPP-503

1 Method of plant cell culture,
2 Preparation of starter culture,
3 Preparation of beverages
4 Production of amylase, flavor, color by fermentation.
5 Immobilization of enzymes.

FPP-504: CEREAL AND PULSE TECHNOLOGY

.**Course Objectives**:
• To create awareness about the processing of major cereals like paddy, maize etc.
• To study the storage and handling techniques of cereals.
• To gain knowledge on processing and milling of pulses.
• To study about the byproducts obtained during processing along with their uses.

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.

- The students are required to attempt one question from each All questions carry equal marks

Unit I

Paddy Processing: Composition of paddy and Quality characteristics. Curing of Paddy – Parboiling Processes. Physico-chemical changes during parboiling – Cold Water soaking and Hot water soaking processes.

Methods of grain drying- LSU Dryer, By Products of Paddy Processing - Paddy husk and its uses as husk ash, activated carbon, furfural and other byproducts. Production of Flattened and Puffed Rice from Paddy.

Unit II

Rice Milling: Paddy Dehusking Processes. Rice Milling Flow Chart. Engelberg Huller Mills. Modern Rice Mills – their Components - Pre Cleaners, Shellers, Under Runner Shellers and Centrifugal Shellers.

Paddy Separators – Satake and Schule Designs, Polishers -Cone polishers and other types, Bran and Brokens separators. Rice Mill yields and loss due to broken at different stages of milling.Use of Rice Bran inEdible oil Industry.

Unit III

Milling Of Pulses: Major Pulses grown in the country .Traditional milling process -merits and demerits. Drying of legumes - Sun and Traditional drying processes steps – Precleaning, Pitting, Oil application, conditioning, dehusking and splitting - Machinery and equipment employed.

Modern milling process - Process flow chart -Mechanical hot air drying and conditioning - merits and demerits, dehusking in Pulse Pearler, Water conditioning, splitting of pulses in Pulse splitter, Merits and demerits. Mini dhal mill -working principle - advantages and disadvantages. Grinding of split pulses, pulse flour products, their applications and equipments used.

Unit IV

Milling And Processing Of Maize: Dry milling of maize, Storage and drying, Pre-cleaning, cleaning equipment, degermination and dehusking, Roller milling, Sifting, Purifying, Aspiration, Pneumatics in a maize mill.

Equipment needed for Degermination, Debraning and starch separation. Starch conversion into other value added products – Acid Hydrolysis, Enzyme Hydrolysis, Isomerization processes. Processing for Dextrose, Malto Dextrin and other products. Extraction and refining of Corn oil in brief. Grain storage and handling

Text Books

1.Chakraverty, A.: Post Harvest Technology of Cereals, Pulses and Oilseeds. Oxford and IBH Publishing Co, Calcutta, 1995.

2. Samuel A .Matz: "The Chemistry and Technology of Cereals as Food and Feed", Chapman and Hall, 1992.

Reference Books

1. Smartt J, "Tropical Pulses", Longman Group Ltd. London, 1980.
2. Pomeranz Y, "Modern Cereal Science and Technology" VCH Publishing Inc. New York, 1987.
3. Cryde M. Christensen, "Storage of Cereal Grains and their Products", American Association of Cereal Chemists inc., St. Paul, USA, 1982.
4. Bernard Godon and Claude Willm, "Primary Processing of Cereals" Berns and Noble Publishers, 1994.
5. KarelKulp and Joseph P Pante, "Handbook of Cereal Science and Technology", MercelDekkar, USA, 2000.

Practicalbased on FPP-504

1. Determination of physical properties of different cereal grains
2. Determination of water absorption capacity ,sedimentation value and alcoholic acidity of the Maida.
3. Determination of adulterant (NaHCO3) in wheat flour/ Maida.
4. Estimation of Protein content of different Cereals and Legumes.
5. Assessment of market samples of wheat, rice, and pulses for conforming to some PFA specifications.
6. Determination of Gluten content in wheat flour samples.
7. Collection of samples at various steps of milling and analysis for efficiency of cleaning, shelling, paddy separation, and degree of polish.

FPP-505:BAKERY AND CONFECTIONERY TECHNOLOGY

Course Objectives:

• To provide knowhow on the machinery and process involved in the baking process
• To understand the various types of sugar and its grades
• To know the confectionery product manufacture

Instructions for paper setters

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt one question from each All questions carry equal marks

Unit I

Bakery Equipments And Testing Of Flour For Bakery Goods: Laboratory testing of Wheat grain Quality, Moisture tests, Grain hardness testing. Testing, Viscograph, Amylograph, Ferinograph. Dough mixers, Dividers, rounders, Proofing, moulding, Ovens, Slicers, Packaging materials and equipment, Sanitation and safety.

.Unit II

Baked Products: Materials of baking, Bread manufacturing process – Straight dough fermentation, Sponge and dough, Biscuit-Types of biscuit dough – Developed dough, short dough, semi-sweet, enzyme modified dough and batters- importance of the consistency of the dough- Cake – Flour specification – ingredients – manufacturing process – types of chemically aerated goods.

Unit III

Sugar Production Process Details: Energy and material balance of cane sugar process. Extraction of juice, extraction yields, drying and uses of Bagasse, Purification of juices-juice filtration and chemical purification, Clarification stages, Lime addition, pH control, Treatment of clarified juice, evaporation –multiple effect evaporators, Vacuum pans, Crystallization, Washing of sugar crystals and centrifugal separation/dewatering of sugar and other related processes. Sugar Refining, Sugar analysis, Sugar recovery –improvement, /Sugar balance, energy conservation, Sugar plant sanitation.

Unit IV

Technology Of Beverages Manufacture (Alcoholic And Non Alcoholic): Manufacture of beer, wine and champagne - Quality characteristics
Manufacture of distilled beverages including whisky, brandy, rum and gin – Quality aspectsManufacture of carbonated beverages – quality aspects – Manufacture of sugar-free, sugarless, carbonated beverages.
Confectionery Technology: Types of Confectionery, raw materials and processing of toffee, chocolates, fruit drops, hard boiled candies. Additives for Confectioneries.Equipments used in Confectionery manufacture.

Text/ Reference Books

1. Samuel A. Matz, “Bakery Technology and Engineering”, Chapman & Hall, 3rd Edition, 1992.
2. “Association of Operative Millers Cereal Millers Hand Book”, Burgess Publishing company, USA, 1963.
3. Pomeranz Y, “Modern Cereal science and Technology” MVCH Publications, NY, 1987.
4. Kent N.L.,.Evers A.D, “Technology of Cereals” Peregaman Press, Elsevier Publishers, 1994.
5. Samuel A. Matz, “Equipment for Bakers” Pan Tech International Publication, 1988.
6. Stanley P Cauvain, Linda S Young, “Technology of Bread making”, Aspen publication, 2ndEdition, 2007.

Practical based onFPP-505

1. Prepare the Bread rolls/cakesand assessment of its quality
2. Perform the quality test on fat/butter for bakery applications on different parameters
3. Carry out the assessment of market Candy and Chewing gum.
4. Preparation of toffee from apple and tomato pulp and to perform its quality assessment tests.

5. Prepare Chocolate cookies and assessment of its quality.

Semester VI

Paper Title: GC 602 - TOTAL QUALITY MANAGEMENT

Objective: The objective of the paper is to provide knowledge to students of the concepts of total quality management and to inculcate among them a concern for quality and customer satisfaction.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Introduction: Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs and its Analysis, Basic concepts of Total Quality Management, Principles of TQM.

UNIT II

Quality Council and Statements, Barriers to TQM Implementation. Quality: Customer Perception and Satisfaction, Customer Complaints, Service Quality, Customer Retention, Continuous Process Improvement – Juran Trilogy, PDSA Cycle, 5S, Kaizen, Supplier Partnership –Selection and Rating

UNIT III

Business Process Reengineering (BPR). Benchmarking –Introduction, Reasons, Process, Quality Function Deployment (QFD) – House of Quality, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM)

UNIT IV

Latest Challenges of Quality, Six Sigma Concepts: application. Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System– Elements, Implementation of Quality System, Documentation, Quality Auditing; ISO 14000, Requirements and Benefits.

Books Recommended:

- 'Total Quality Management' by A.V. Feigenbaum, McGraw-Hill Publication.
- 'Total Quality Management' by J.S. Oakland, Butterworth – Hcinemann Ltd., Oxford Press.
- 'Quality Management – Concepts and Tasks' by V. Narayana & N.S.Sreenivasan, New Age International.
- Total Quality Management for Engineers' by Zeiri, Wood Head Publishers.
- 'Total Quality Management' by James R. Evans, Cengage Learning.

FPP-603: Industrial Training

Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce.These credits will be evaluated in Semester VI.

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